

2938

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4. (a) Explain the implantation of embryo in human. (6)
- (b) Define teratogenesis, Discuss pharmaceutical drugs and alcohol as teratogens. (6)
5. (a) What are the different extra embryonic membranes in birds. Add a note on their evolutionary significance. (4)
- (b) Describe various morphogenetic movements with reference to gastrulation in chick. (8)
6. (a) Explain the process of neural tube formation with suitable diagrams. Name a neural tube defect which occur due to improper closing of anterior neuropore. (6)
- (b) 'The amount and deposition of yolk affects the cleavage pattern' Justify the statement. (6)
7. Write short notes on any **three** of the following : (3×4=12)
- (a) Amniocentesis
- (b) Types of placenta on the basis of histology
- (c) Techniques to study fate maps
- (d) Cortical Reaction
- (e) Embryonic stem cell

(1000)

28.5.24 (M)

[This question paper contains 4 printed pages]

Your Roll No.

Sr. No. of Question Paper : 2938

Unique Paper Code : 32231601

Name of the Paper : Developmental Biology

Name of the Course : B.Sc. (H) Zoology

Semester : VI (LOCF)

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
 2. Attempt **Five** questions in all. Question No. 1 is compulsory.
 3. Attempt **all** the parts of question together.
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1. (a) Define the following (**any five**) : (5)
 - (i) Invagination
 - (ii) Amphimixis
 - (iii) Archenteron
 - (iv) Morphogen
 - (v) Window of implantation

P.T.O.

(b) Distinguish between any **four** of the following :
(2×4=8)

- (i) Splanchnopleure and Somatopleure
- (ii) Coeloblastula and Steroblastula
- (iii) Progressive and Retrogressive metamorphosis
- (iv) Area Opaca and Area Pellucida
- (v) Epimorphosis and Morphallaxis

(c) Match the following : (3)

- | | |
|--------------------------|----------------------------|
| (i) E. Allen | a. Phocomelia |
| (ii) Thalidomide | b. DDT as teratogen |
| (iii) A. Trembley | c. Senescence |
| (iv) Rachel Carson | d. Superovulation |
| (v) Clomiphene | e. Amphibian metamorphosis |
| (vi) Genetic instability | f. Hydra Regeneration |

(d) Expand the following abbreviations : (3)

- (i) IVF
- (ii) GGF
- (iii) DLHP

(e) Fill in the blanks : (4)

- (i) Second meiotic arrest in human oocyte takes place at _____ stage.
- (ii) Absence of _____ makes the sperm immotile.
- (iii) Chemoattractants identified in the egg jelly of sea urchin are _____ and _____.

(f) Name the germ layer from which each of the following is derived : (4)

- (i) Trachea
- (ii) Ovary
- (iii) Bone
- (iv) Adenohypophysis

2. (a) Differentiate between Holometabolous and Hemimetabolous development. Discuss the role of hormones in insect metamorphosis. (6)

(b) What are the various causes of ageing, describe any two in detail. (6)

3. (a) Discuss the process by which the spermatid transforms into spermatozoa. (6)

(b) What do you understand by embryonic induction. Write the experimental proof of the regional specificity of induction. (6)